

Why UHD restoration is hard

- Downsample $\rightarrow$ enhance $\rightarrow$ upsample destroys 4K detail for good.
- Pure frequency-domain models discard the *locality* of a degradation.
- Retinex methods are low-light only, and give reflectance and illumination one shared architecture.

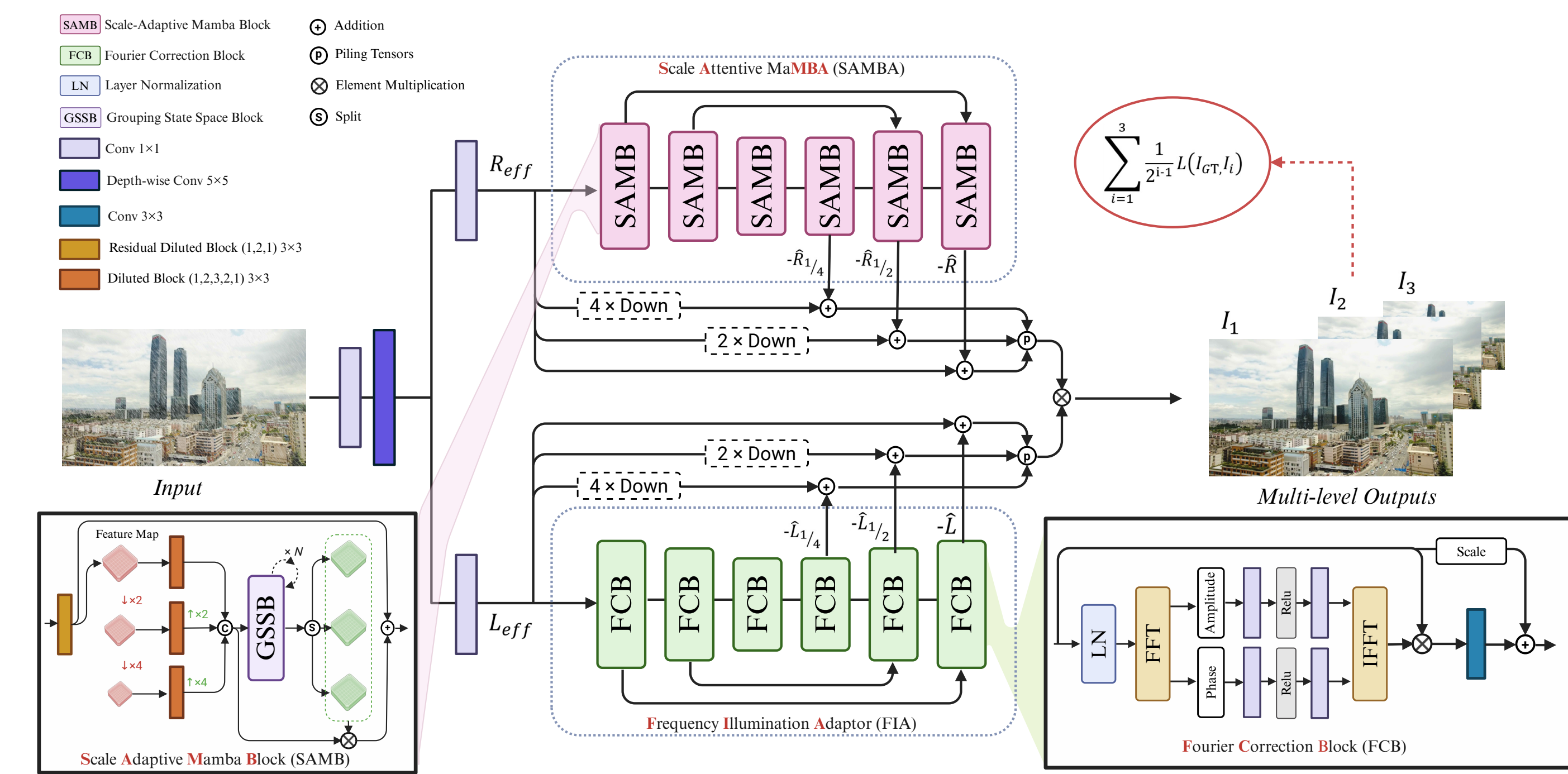
Key finding

The degradation decides the domain — no single-domain model generalizes.

- Global haze:** Fourier wins by **7.3 dB**.
- Local blur:** spatial wins by **6.5 dB**.

So we split the image by *component*, not by task.

Method



SAMBA (spatial, Mamba) repairs reflectance; **FIA** (Fourier) repairs illumination. Three scales are recombined under Retinex at every level and deep-supervised.

Retinex as a general prior

Retinex separates what an object *is* from how it is *lit*; degradation perturbs both, and each is repaired separately.

$$I = R_{\text{eff}} \odot L_{\text{eff}} = (R + \hat{R}) \odot (L + \hat{L})$$

$$\hat{R} = -S(R_{\text{eff}}) \text{ (SAMBA)}, \quad \hat{L} = -\mathcal{F}(L_{\text{eff}}) \text{ (FIA)}$$

Frequency analysis: which domain sees the degradation?



PSNR measured *within* each domain — higher means the degradation is compactly described there. Haze is compact in Fourier, blur in space.

Inside the two branches

SAMBA • reflectance

Mamba scans causally — pixel i sees only pixels $< i$. SAMBA scans three scales at once so distant regions become neighbours; GSSM adds an embedding, $y = (C+E)h + Dx$.

$$x_s = \text{RDB}(x_{in})$$

$$x_0, x_1, x_2 = \text{Up}(\text{DB}(x_s, x_{s\downarrow}, x_{s\downarrow\downarrow}))$$

$$\hat{x}_0, \hat{x}_1, \hat{x}_2 = \mathcal{S}(\text{GSSB}(\mathcal{C}[x_0, x_1, x_2]))$$

$$x' = \sum_{i=0}^2 x_i \odot \hat{x}_i + x_s$$

Training objective

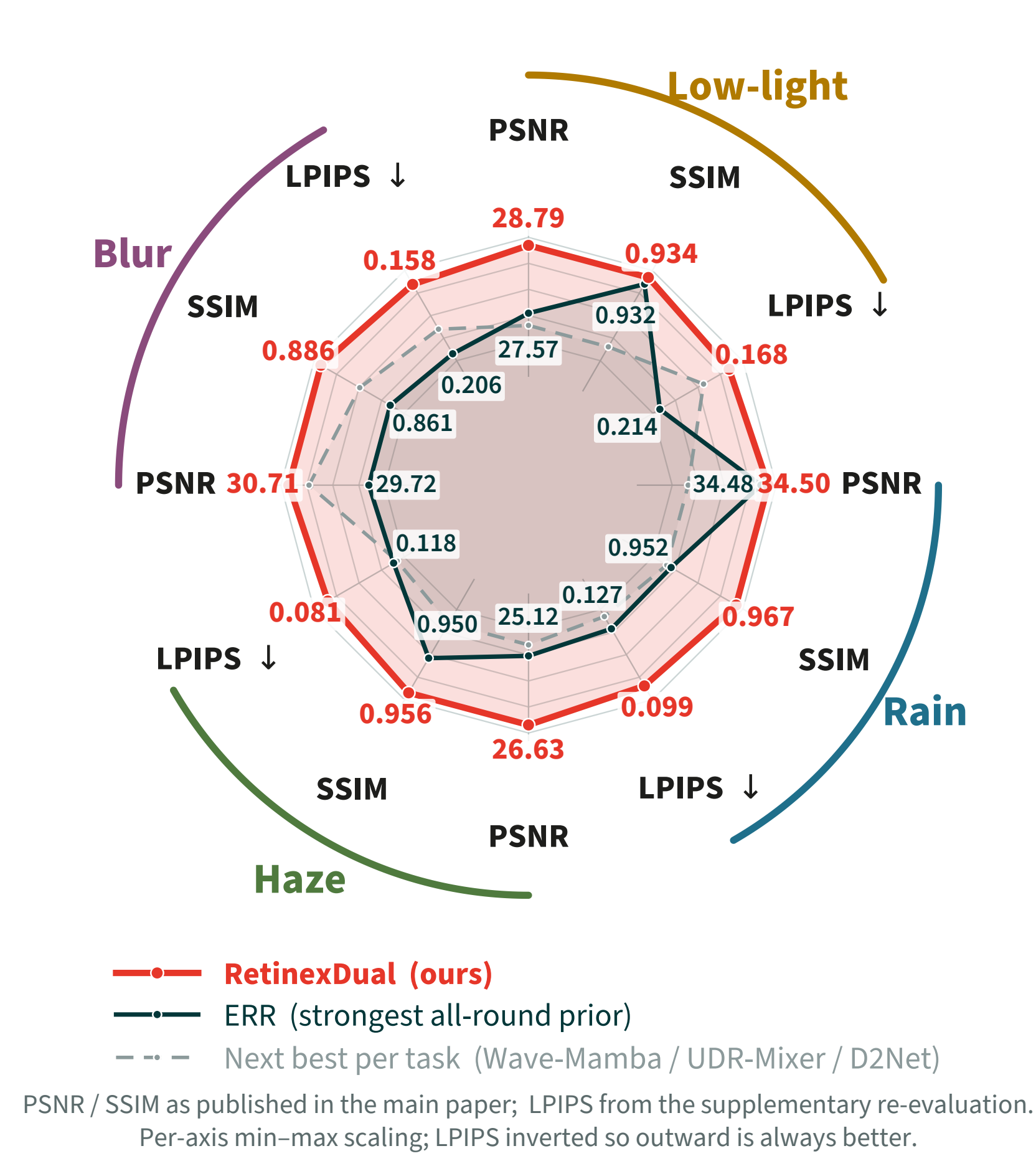
Four complementary terms, weighted 1, 0.1, 0.5, 0.4, and deep-supervised at three scales — each level reconstructed through Retinex so the decomposition is never violated.

$$\mathcal{L} = \lambda_{cb}\mathcal{L}_{cb} + \lambda_{\text{FFT}}\mathcal{L}_1(\text{FFT}) + \lambda_{ssim}\mathcal{L}_{ssim} + \lambda_p\mathcal{L}_p, \quad \mathcal{L}_{\text{multi}} = \sum_{i=1}^3 \frac{1}{2^{i-1}} \mathcal{L}(l_i, \hat{l}_i)$$

Contributions:

- First Retinex framework for **general** UHD restoration.
- SAMBA**: Mamba scanned over three feature scales.
- FCB**: Fourier colour correction in **0.2 M** parameters.

Results — best bold, previous best underlined



UHD-LL • Low-light

Method	PSNR	SSIM	Par.
Restormer	22.25	0.871	26.1M
SNR-Aware	22.72	0.877	40.1M
LLFormer	22.79	0.853	13.2M
UHDFour	26.23	0.900	17.5M
UHDFormer	27.11	0.927	0.3M
Wave-Mamba	27.35	0.913	1.2M
ERR	<u>27.57</u>	<u>0.932</u>	1.1M
Ours	28.79	0.934	4.7M

4K-Rain13K • Rain

Method	PSNR	SSIM	Par.
SPDNet	31.81	0.922	3.0M
IDT	32.91	0.948	16.4M
DRSformer	32.94	0.933	33.6M
Restormer	33.02	0.933	26.1M
UDR-S2Former	33.36	0.946	8.5M
UDR-Mixer	34.28	0.951	8.5M
ERR	<u>34.48</u>	<u>0.952</u>	1.1M
Ours	34.50	0.967	4.7M

UHD-Haze • Haze

Method	PSNR	SSIM	Par.
Restormer	12.72	0.693	26.1M
DehazeFormer	15.37	0.725	2.5M
UHD	18.04	0.811	34.5M
Uformer	19.83	0.737	20.6M
UHDFormer	22.59	0.943	0.3M
D2Net	24.88	0.944	5.2M
ERR	<u>25.12</u>	<u>0.950</u>	1.1M
Ours	26.63	0.956	4.7M

UHD-Blur • Blur

Method	PSNR	SSIM	Par.
Stripformer	25.05	0.750	19.7M
Restormer	25.21	0.752	26.1M
Uformer	25.27	0.752	20.6M
FFTformer	25.41	0.757	16.6M
UHDFormer	28.82	0.844	0.3M
ERR	29.72	0.861	1.1M
D2Net	<u>30.46</u>	<u>0.872</u>	5.2M
Ours	30.71	0.886	4.7M

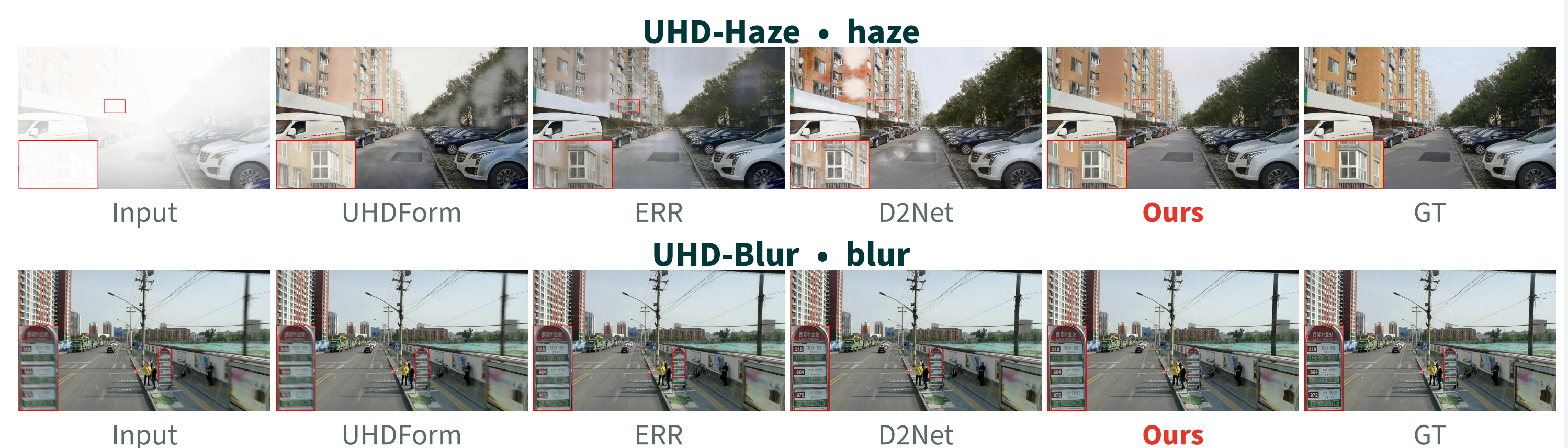
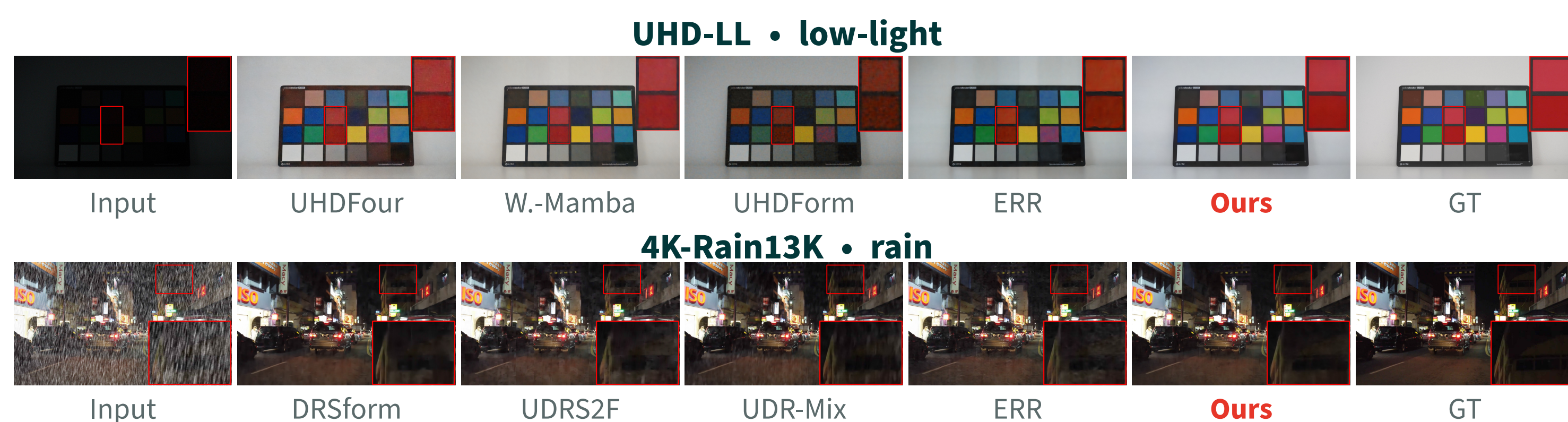
Ablation • design

Variant	PSNR	SSIM	Par.
<i>Retinex duality</i>			
FIA for both R, L	24.83	0.866	0.42M
SAMBA for both R, L	27.91	0.920	9.03M
w/o FIA branch	26.67	0.919	4.52M
w/o SAMBA branch	23.83	0.866	0.21M
<i>Components</i>			
w/o multi-scale	26.96	0.906	1.34M
w/o GSSB	25.49	0.908	1.70M
w/o Fourier (FCB)	27.03	0.919	4.68M
Full model	28.79	0.934	4.73M

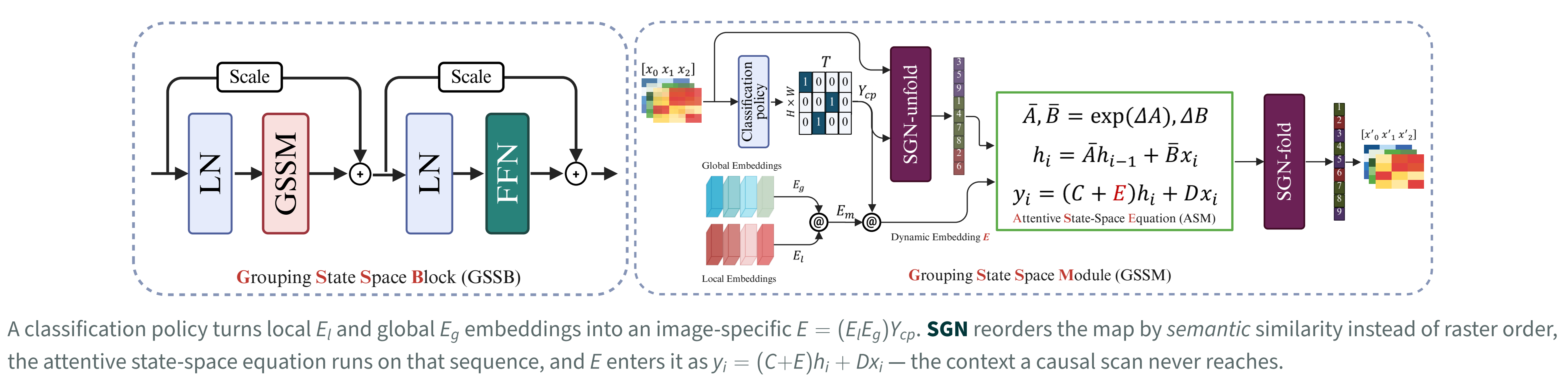
Ablation • objective

Configuration	PSNR	SSIM
w/o $\mathcal{L}_{cb}$	27.27	0.911
w/o $\mathcal{L}_{\text{FFT}}$	27.92	0.918
w/o $\mathcal{L}_{ssim}$	27.15	0.906
w/o $\mathcal{L}_p$	27.20	0.911
w/o multi-level	26.63	0.906
w/o level scaling	28.54	0.921
Full objective	28.79	0.934

Qualitative comparison — every baseline shown in the paper's own comparison figures



Group State Space Module — inside every SAMBA



A classification policy turns local E_l and global E_g embeddings into an image-specific $E = (E_l E_g) Y_{cp}$. **SGN** reorders the map by *semantic* similarity instead of raster order, the attentive state-space equation runs on that sequence, and E enters it as $y_i = (C+E)h_i + Dx_i$ — the context a causal scan never reaches.

Conclusion

- Retinex is a viable prior for **general** UHD restoration, not just low-light: one framework, SOTA on **four** degradations.
- The gain comes from **asymmetry** — multi-scale Mamba for reflectance, a compact Fourier block for illumination.
- Limitation:** at 0.955 s and 4.7 M parameters the model is not optimized for speed or size.



Code & models

github.com/ErrorLogic1211/RetinexDual

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